Ain Shams University
Faculty of Engineering
Electronics and Electrical Communication Department
ECE 433s Microwave Circuits and Systems
Spring 2026

Design of a microwave amplifier

Students are expected to go through a full real-world engineering design cycle for developing a microwave amplifier circuit. This cycle includes:

- Preliminary pencil and paper design work.
- Simulation and optimization of the **realistic** implementation of the circuit using ADS microwave simulator.
- Preparation of the layout necessary for the fabrication of the circuit.

Project Substrate

The circuits will be fabricated using **microstrip** transmission lines. The substrates are a high-quality, low-loss microwave substrate that has the following characteristics:

- **Type 1: Rogers RT/Duroid 5880**
  - Thickness: 787.41 µm.
  - Dielectric constant: $\varepsilon_r = 2.20 \pm 0.04$.
  - Dielectric losses: $\tan\delta = 0.0009$ at 10 GHz.
  - Copper thickness: 1 ounce per square foot or 34 µm.
- **Type 2: Rogers 6010**
  - Thickness: 635 µm.
  - Dielectric constant: $\varepsilon_r = 10.2 \pm 0.25$.
  - Dielectric losses: $\tan\delta = 0.0025$ at 10 GHz.
  - Copper thickness: 1 ounce per square foot or 34 µm.

Project Specifications

A microwave amplifier will be designed, and simulated at your assigned frequency in a 50 Ω system. The s-parameters file for npn type BJT BFG410W fabricated by NXP semiconductors will be delivered to the students. The following design specifications must be met:

- Gain: $G_{T,max}$ at the design frequency.
- Stability: stable over the entire useful bandwidth of the device.
- $S_{11}|_{amp}$: minimize.
- $S_{22}|_{amp}$: minimize.
- Bandwidth: maximize, based on a narrowband design.

Single stub matching networks are to be designed and constructed using distributed elements (T.L.'s). Only open-circuited shunt stubs should be used. All possible matching networks should be designed, simulated, and optimised. The pair of networks that are to be used in the end should be selected such that the bandwidth of the amplifier is maximized. Each type of network has a different frequency response, and the frequency response of the amp will depend on the combination of matching networks and on the device. You may do the paper design assuming lossless lines but should account for losses in the simulation.

The transistors will be biased using two dc power supplies; you must design bias tee's (RF choke and dc block) and incorporate them at the input and output of your amplifier. The RF choke will be designed based on transmission line techniques. It consists of radial stub and λ/4 transmission line. Your amplifier must be dc isolated by a blocking capacitors. **You should keep in mind that λ is the waveguide wavelength, which is $\lambda_o/\sqrt{\varepsilon_{eff}}$, where ε_{eff} is the effective permittivity of the microstrip line.** The transistor biasing circuit and the RF choke design are illustrated in a separate PDF file.

It is important that the physical dimensions of all external elements (transistor, C, R) be taken into account in your design. The physical description of the package is given in the specification sheets provided. The emitter pads of the transistor must be big enough to be drilled in order to be connected to the ground plane via a thin wire.

Simulation and layout Work

The design of all circuits must be done on paper first and then simulated, refined, and optimized using ADS. This is a popular microwave circuit simulator that can be used as a design aid to obtain the theoretical performance of a realistic microwave circuit before fabrication. The transistor is assumed to be a linear two-port network having distinct values of S-parameters at different frequencies. In your DC bias design, you will select the proper values of resistors to have **I_c=6 mA and V_{ce}=2 V**. ADS has the capability to generate the final circuit layout.

Project organization

This project accounts for 15 percent of the final grade in the course. Students must work in groups of three. Each group will work at an assigned central frequency and a substrate. **The due date for submitting the reports is May 1, 2026.**

Check your layout so that:

1- You have space between the launcher microstrip lines and the surface mount capacitors.
2- Your layout must take into consideration the physical dimensions of the transistor, capacitors, and resistors. The capacitors and resistors are surface-mounted ones. You can use the dimensions of the 0603 package for the capacitors and resistors.

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Group number	Student name	Center frequency (GHz)	Substrate type.
1		4	Rogers 6010
2		4.5	Rogers 6010
3		4	Rogers RT/Duroid 5880
4		4.5	Rogers RT/Duroid 5880
5		5	Rogers RT/Duroid 5880
6		5	Rogers 6010
7		5.5	Rogers RT/Duroid 5880
8		6	Rogers RT/Duroid 5880
9		5.5	Rogers 6010
10		6	Rogers 6010
11		6.5	Rogers RT/Duroid 5880
12		6.5	Rogers 6010
13		7	Rogers RT/Duroid 5880
14		7	Rogers 6010
15		7.5	Rogers 6010
16		8	Rogers 6010
17		8.5	Rogers 6010
18		7.5	Rogers RT/Duroid 5880
19		8	Rogers RT/Duroid 5880
20		8.5	Rogers RT/Duroid 5880
21		9	Rogers 6010
22		9	Rogers RT/Duroid 5880
23		9.5	Rogers 6010
24		9.5	Rogers RT/Duroid 5880
25		10	Rogers RT/Duroid 5880
26		10	Rogers 6010

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Evaluation criteria

Hand analysis of the microwave amplifier	5
Simulation window showing all components. Explanation of each figure.	2
Simulation figures	4
Calculating the bandwidth based on simulation results	2
Neatness	2
Total	15